



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

ELECTRICAL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Electrical Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers.

QUESTION 1

1 1.1 C ✓✓ 1.2 B ✓✓ 1.3 B ✓✓ 1.4 B ✓✓ 1.5 B ✓✓ [10]

QUESTION 2

- 2.1** Any THREE: touching live (exposed) wires ✓; damaged insulation on cables ✓; working on circuits that are switched on ✓; wet hands or wet floors; faulty appliances without earthing (3)
- 2.2** Do not touch the person ✓. Switch off the power at the main switch or remove the plug, or push the person away with a dry non-conducting object (wood) ✓. Call for medical help and give first aid / CPR if trained ✓. (3)
- 2.3** It detects current leaking to earth (e.g. through a person) ✓ and switches off the supply quickly to prevent electric shock and fire ✓. (2)
- 2.4** Any FOUR: never work on live circuits ✓; use insulated tools ✓; keep the work area dry ✓; get the teacher to check a circuit before switching on ✓; know where the main switch and fire extinguisher are; no horseplay; report damaged equipment (4)

[12]

QUESTION 3

- 3.1** Side cutters: cut wires ✓; long-nose pliers: hold and bend small wires and parts in tight spaces ✓; wire strippers: remove insulation from wires ✓; soldering iron: melt solder to join components and wires ✓ (4)
- 3.2** In series with the load ✓ (the circuit must be broken to insert the meter), with the dial on the current (A) range ✓. Precaution, any ONE: start on the highest range / check polarity / never connect it across the supply ✓. (3)
- 3.3** Any THREE: shiny and smooth ✓; concave (volcano) shape ✓; solder flows around the whole joint ✓; no dry joints or cracks; not too much solder (3)
- 3.4** Yellow 4, violet 7 ✓, orange $\times 10^3$ ✓ = 47 000 Ω = 47 k Ω , gold $\pm 5\%$ ✓ (3)

[13]

QUESTION 4

- 4.1** The current through a conductor is directly proportional to the potential difference across it ✓, provided the temperature stays constant ✓. (2)
- 4.2.1** $R_T = 10 + 20 + 30 \checkmark = 60 \Omega \checkmark$ (2)
- 4.2.2** $I = \frac{V}{R} = \frac{12}{60} \checkmark\checkmark = 0,2 \text{ A } \checkmark$ (3)

- 4.2.3 $V = IR = 0,2 \times 20 \checkmark\checkmark = 4 \text{ V } \checkmark$ (3)
- 4.2.4 $P = VI = 12 \times 0,2 \checkmark\checkmark = 2,4 \text{ W } \checkmark$ (3)
- 4.3 $\frac{1}{R} = \frac{1}{10} + \frac{1}{10} \checkmark \rightarrow R = 5 \Omega \checkmark$ (2)
- 4.4 The total current entering a junction equals the total current leaving the junction $\checkmark\checkmark$. (2)
- 4.5 (a) 3 V $\checkmark$ (b) 1,5 V $\checkmark$. Advantage of parallel: the cells last longer / can supply more current at the same voltage $\checkmark\checkmark$. (4)
- 4.6 Any TWO, 2 marks each: DC flows in one direction only, while AC changes direction continuously $\checkmark\checkmark$; DC comes from cells and batteries, while AC comes from generators / the mains (230 V, 50 Hz) $\checkmark\checkmark$; AC voltage can easily be changed with a transformer. (4)

[25]

QUESTION 5

- 5.1 Forward bias: the anode is connected to the positive and the cathode to the negative of the supply $\checkmark\checkmark$. The diode conducts once the voltage is above about 0,7 V (silicon) $\checkmark$, so current flows $\checkmark$. (4)
- 5.2 To limit the current $\checkmark$ so that the LED is not damaged (burnt out) $\checkmark$. (2)
- 5.3 $V_R = 9 - 2 = 7 \text{ V } \checkmark$; $R = \frac{7}{0,02} \checkmark = 350 \Omega \checkmark$ (3)
- 5.4 It stores electric charge (energy) $\checkmark$ and releases it when needed (e.g. smoothing, timing) $\checkmark$. (2)
- 5.5 As the light gets brighter, its resistance decreases $\checkmark$; in the dark, the resistance increases $\checkmark$. Use, any ONE: automatic street lights / night lights / light meters $\checkmark$. (3)
- 5.6 Any TWO: single-pole single-throw (SPST) $\checkmark$; single-pole double-throw (SPDT) $\checkmark$; push-button; toggle; slide; rotary; reed switch (2)
- 5.7 A fuse melts and must be replaced after it blows $\checkmark\checkmark$, while a circuit breaker trips and can be reset. (2)
- 5.8 Its resistance decreases $\checkmark\checkmark$. (2)

[20]

QUESTION 6

- 6.1 $13 = 8 + 4 + 1 \checkmark = 1101_2 \checkmark$ (2)
- 6.2 $16 + 4 + 2 \checkmark = 22 \checkmark$ (2)

6.3

AND			OR		
A	B	Y	A	B	Y
0	0	0	0	0	0
0	1	0	0	1	1
1	0	0	1	0	1
1	1	1	1	1	1

(6)

Each table: headings ✓, all input combinations ✓, correct outputs ✓

6.4

It inverts the input ✓: an input of 1 gives an output of 0, and 0 gives 1 ✓.

(2)

6.5

AND: $Y = A \cdot B$ ✓; OR: $Y = A + B$ ✓

(2)

6.6

NAND gate ✓.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

✓✓✓

(4)

6.7

Any TWO: electronic circuits easily represent two states (on/off, high/low voltage) ✓;
binary signals are less affected by noise / more reliable ✓; simple and cheap to build
circuits

(2)

[20]

TOTAL: 100

END OF MARKING GUIDELINE